5 Motion in a Plane

General Motion in 2D

1. If initial velocity of object $\vec{u} = 3\hat{i} + 4\hat{j}$ and after some time its velocity $v = 4\hat{i} + 3\hat{j}$ then find
   (*i*) magnitude of change in velocity
   (*ii*) change in magnitude of velocity.

2. Position of object $\vec{r} = 3t^2\hat{i} + (6t - t^2)\hat{j}$, then find velocity and acceleration at $t = 1$ sec.

3. x and y -coordinates of the particle at any time are $x = 5t - 2t^2$ and $y = 10t$, where x and y. Acceleration of the particle at $t = 2$s.

4. A particle has initial velocity $2\hat{i} + 3\hat{j}$ and acceleration $(0.3\hat{i} + 0.2\hat{j})$. Magnitude of velocity after 10 sec.

5. A particle moving in plane such that $y = \frac{x^2}{4}$ and $x = \frac{t}{2}$ then find speed at $t = 2$ sec.

6. A particle starting from origin moves in straight line and reach at a point $(\sqrt{3}, 3)$ then path makes an angle.

7. Object starts from the point $(2\hat{i}, 4\hat{j})$m at $t = 0$ with velocity $(5\hat{i} + 4\hat{j})$ with constant acceleration $(4\hat{i} + 4\hat{j})$m/s^2. What is distance of particle from origin at $t = 2$ sec.

8. Potion of object $\vec{r} = (t^2 - 38t)\hat{i} + 2t^3\hat{j}$ find instant when velocity and acceleration are perpendicular.

9. A car starts from rest and acceleration 5m/s^2 at $t = 4$ s. A ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball at $t = 6$s?
   (*a*) $20\sqrt{2}$ m/s,10 m/s^2, $t = 0$ (*b*) 20 m/s, 5 m/s^2
   (*c*) 20 m/s,0 (*d*) $20\sqrt{2}$m/s,0

10. A particle starts from origin with velocity $3\hat{i}$ m/s and acceleration $(6\hat{i} + 4\hat{j})$. Then find x-coordinates of particle when y -coordinates is 32.

11. Object is moving with velocity $V = 3\sin(\omega t)\hat{i} + 3\cos(\omega t)\hat{j}$ then find distance moved by object in 2 sec.

12. At time $t = 0$ a particle starts travelling from a height $7\hat{z}$ cm in a plane keeping z coordinate constant. At any instant of time it's position along the $\hat{x}$ and $\hat{y}$ directions are defined as $3t$ and $5t^3$ respectively. At $t = 1$s acceleration of the particle will be
   (*a*) $-30\hat{y}$
   (*b*) $30\hat{y}$
   (*c*) $3\hat{x} + 15\hat{y}$
   (*d*) $3\hat{x} + 15\hat{y} + 7\hat{z}$

13. Acceleration of object $a = 2\hat{i} + 3t^2\hat{j}$, then find velocity at $t = 1$ sec it initial velocity of object is zero.

14. Initial velocity of object $(4\hat{i} + 8\hat{j})$m/s and acceleration $a = -4$ m/s$^2\hat{j}$, then find velocity after $t = 2$sec and displacement in 3 sec.

15. Match column-I with Column-II:

Column-I (Condition)		Column-II (Result)	
(A)	$\vec{v} \cdot \vec{a} = +ve$	(*i*)	Velocity constant
(B)	$\vec{v} \cdot \vec{a} = -ve$	(*ii*)	Speed constant
(C)	$\vec{v} \cdot \vec{a} = 0$	(*iii*)	Speed increases
		(*iv*)	Speed decrease

16. **Assertion (A):** If acceleration is present, it will surely change speed of the object.
   Reason (R): Acceleration is rate of change in velocity.
   (*a*) Assertion (A) is false but Reason (R) is true
   (*b*) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
   (*c*) Assertion (A) is true but Reason (R) is false
   (*d*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

17. For initial velocity $(\vec{v}_i) = 3\hat{i} - 4\hat{j}$ and find velocity $(\vec{v}_f) = 3\hat{i} - 4\hat{j}$

(A)	$\lvert\Delta\vec{v}\rvert$	(i)	$8\hat{j}$
(B)	$\Delta\lvert\vec{v}\rvert$	(ii)	$6\hat{i}$
(C)	$\vec{v}_f - \vec{v}_i$	(iii)	0
(D)	$\vec{v}_f + \vec{v}_i$	(iv)	8

(*a*) (A)-(iii); (B)-(iv); (C)-(ii); (D)-(i)

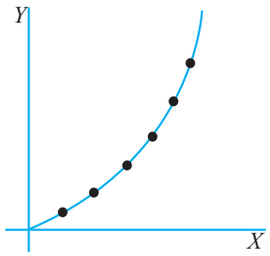
(*b*) (A)-(iii); (B)-(iv); (C)-(i); (D)-(ii)

(*c*) (A)-(iv); (B)-(iii); (C)-(ii); (D)-(i)

(*d*) (A)-(iv); (B)-(ii); (C)-(i); (D)-(ii)

Equation of Trajectory

18. Position of object at time $'t', \vec{r} = 2t\hat{i} + 4t^2\hat{j}$, then find equation of trajectory.



19. If position of object $\vec{r} = 3\sin(\omega t)\hat{i} + 3\cos(\omega t)\hat{j}$, then object is moving on.

20. If position of object $\vec{r} = 4\sin(\omega t)i + 3\cos(\omega t)\hat{j}$, then object is moving on

21. If position of object $\vec{r} = 4\sin(\omega t)\hat{i} + 3\sin(\omega t)\hat{j}$, then object is moving on

(*a*) ellipse (*b*) circle

(*c*) parallel (*d*) straight line

22. Velocity of object $\vec{V} = 2\hat{i} + x\hat{j}$ then equation of trajectory.

23. A particle moving with velocity $V = y\hat{i} + x\hat{j}$ then find equation of trajectory.

Ground to Ground Projectile Motion

24. A bullet is fired from a gun at the speed of 280 ms^{-1} in the direction 30° above the horizontal. The maximum height attained by the bullet is... (g = 10 m/s^2, sin 30° = 0.5)

(*a*) 2000 m (*b*) 1000 m

(*c*) 3000 m (*d*) 2800 m

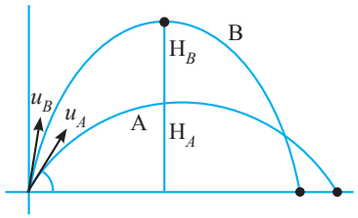
25. Velocity of projection $\vec{u} = 40\hat{i} + 30\hat{j}$ then find H, T, R and angle of project.

26. The horizontal range and the maximum height of a projectile are equal. The angle of projection of the projectile

(*a*) $\theta = \tan^{-1}$ (*b*) $\theta = 45°$

(*c*) $\theta = \tan^{-1}\left(\frac{1}{4}\right)$ (*d*) $\theta = \tan^{-1}(4)$

27. Find the relation between time of flight from a given path as shown in fig.



(*a*) $T_A = T_B$ (*b*) $T_A > T_B$

(*c*) $T_A < T_B$ (*d*) Can't Say

28. The range of a projectile when fired at with the horizontal 75° is 1km what will be it's range when fired at 15° with same speed :-

(*a*) 0.5 km (*b*) 1.0 km (*c*) 1.5 km (*d*) 2.0 km

29. An arrow is shot into the air. Its range is 200 metres and its time of flight is 5 s. If the value of *g* is assumed to be $10ms^{-2}$ then the horizontal component of the velocity of arrow is :-

(*a*) 25 m /s (*b*) 40 m /s

(*c*) 31.25 m/s (*d*) 12.5 m/s

30. Two projectiles of same mass and with same velocity are thrown at an angle 60° & 30° with the horizontal, then which quantity will remain same:

(*a*) Time of flight

(*b*) Horizontal range of projectile

(*c*) Max height acquired

(*d*) All of them

31. For angles of projection of a projectile at angles $(45^\circ - \theta)$ and $(45^\circ + \theta)$, the horizontal ranges described by the projectile are in the ratio of :

(*a*) 1 : 1 (*b*) 2 : 3 (*c*) 1 : 2 (*d*) 2 : 1

32. A projectile is thrown into space so as to have the maximum possible horizontal range equal to 400 m. Taking the point of projection as the origin, the coordinates of the point where the velocity of the projectile is minimum are :-

(*a*) (400, 100) (*b*) (200, 100)

(*c*) (400, 200) (*d*) (200, 200)

33. Two stones are projected with the same speed but making different angles with the horizontal. Their ranges are equal. If the angle of projection of one is π/3 and its maximum height is y_1 then the maximum height of the other will be :

(*a*) $3y_1$ (*b*) $2y_1$ (*c*) $y_1/2$ (*d*) $y_1/3$

34. A projectile has same range at two different angle if time of flight are t_1 and t_2 then range of the projectile will be?

35. The horizontal range of a projectile is $4\sqrt{3}$ times its maximum height. Its angle of projectile will be:

(*a*) 45° (*b*) 60° (*c*) 90° (*d*) 30°

36. Two bodies are thrown up at angles of 45° and 60° respectively, with the horizontal. If both bodies attain same vertical height, then the ratio of velocities with which these are thrown is :

(*a*) $\sqrt{\frac{2}{3}}$ (*b*) $\frac{2}{\sqrt{3}}$ (*c*) $\sqrt{\frac{3}{2}}$ (*d*) $4\frac{\sqrt{3}}{2}$

37. A projectile is fired from the surface of the earth with a velocity of 5 ms^{-1} and angle θ with the horizontal. Another projectile fired from another planet with a velocity of $3ms^{-1}$ at the same angle follows a trajectory which is identical with the trajectory of the projectile fired from the earth. The value of the acceleration due to gravity on the planet is (in ms^{-2}) is: (given g = 9.8 $m.s^2$)

(*a*) 3.5 (*b*) 5.9 (*c*) 16.3 (*d*) 110.8

38. The equation of projectile is $y = \sqrt{3}x - \frac{gx^2}{2}$ the angle of projection is :

(*a*) 30° (*b*) 60° (*c*) 45° (*d*) none

39. The equation of projectile is $y = 16x - \frac{x^2}{4}$ the horizontal range is

(*a*) 16 m (*b*) 8 m

(*c*) 64 m (*d*) 12.8 m

40. The equation of trajectory of a projectile thrown from a level ground near the surface of earth is given by $y = ax - bx^2$ with *y*-axis in vertical direction and *x*-axis in horizontal direction, *a* and *b* are constants. Then,

(*i*) The range of the projectile is a / b.

(*ii*) At $x - a/2b$, the velocity of projectile becomes zero.

(*iii*) The maximum height attained by projectile is $a^2/4b$.

(*iv*) The angle of projectile is $\tan^{-1}(a)$

(*a*) (*i*) and (*ii*) (*b*) (*ii*) and (*iii*)

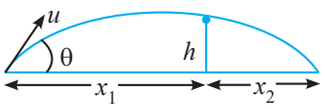
(*c*) (*i*), (*iii*) and (*iv*) (*d*) All above

41. Position of projected object at time *t* is $\vec{s} = 30t\hat{i} + (40t - 5t)\hat{j}$ then find (i) Angle of Projection (ii) H_{max} (iii) *R* & T_f

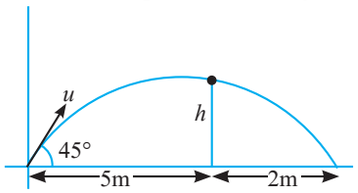
42. Particle is projected and its velocity at time 't' is $\vec{v} = \alpha\hat{i} + (\beta - rt)\hat{j}$ then find H,R,T angle of projection.

43. Particle is projected at other planets & its position at time 't' is $\vec{r} = 3t\hat{i} + (4t - 2t^2)\hat{j}$ find t_f(Time of flight), H (maximum height) and R (range)

44. Find relation between x_1, x_2 and *h*; if the ball is projected with speed *u* at angle θ.



45. A projectile is launched at an angle of 45° with an initial velocity *u* from the ground. The horizontal distance traveled by the projectile is 7 meters (5 meters to the highest point and 2 meters from the highest point to the landing point). Find the maximum height *h* attained by the projectile.



46. Ball is projected, with speed u at angle θ then find time after which ball will be moving perpendicular to the initial direction of projection.

47. A ball is thrown vertically upwards from the top of a cliff. Take starting position of motion as origin and upward direction as positive. Column-I specifies the position, velocity and for acceleration of the particle at any instant column-II gives their sign (+) or (–) at that moment.

Column – I		Column – II	
i.	When the ball is above point of projection, its displacement is	A	Always (+)
ii.	When the ball is above point of projection, its velocity is	B.	Always (–)
iii.	When the ball is above point of projection, its acceleration is	C.	May be (+)
iv.	When the ball is below point of projection, its acceleration is	D.	May be zero

(*a*) (i)–A; (ii)–C, D; (iii) –B; (iv)–B

(*b*) (i)–B; (ii)–A, (iii) –D; (iv)–C

(*c*) (i)–C; (ii)–B, (iii) –C, D; (iv)–B

(*d*) (i)–A; (ii)–C, (iii) –C; (iv)–B

48. The path of projectile is represented by $y = Px - Qx^2$.

Column – I		Column – II	
i.	Range	A	P/Q
ii.	Maximum height	B.	P
iii.	Time of flight	C.	$P^2/4Q$
iv.	Tangent of angle of projection is	D.	$\sqrt{\frac{2P}{Qg}}$

(*a*) (i)–D, (ii)–A, (iii) –B, (iv)–C

(*b*) (i)–C, (ii)–A, (iii) –D; (iv)–B

(*c*) (i)–A, (ii)–C, (iii) –D, (iv)–B

(*d*) (i)–B; (ii)–C, (iii) –A; (iv)–C

49. At maximum height of projectile motion velocity is minimum and perpendicular to acceleration.

(*a*) True (*b*) False

50. Tangential acceleration in projectile motion 1st decreases then increases during motion.

(*a*) True (*b*) False

51. The rate of change in velocity is constant in projectile motion.

(*a*) True (*b*) False

52. Speed, velocity and acceleration are variable in projectile motion.

(*a*) True (*b*) False

53. If air friction is not ignored in projectile motion angle of projection is less than angle of collision.

(*a*) True (*b*) False

54. Ball is projected at angle θ then total angular displacement of ball is 2θ in time of flight of Projectile motion.

(*a*) True (*b*) False

55. Projectile motion is non-uniform motion with non-uniform acceleration.

(*a*) True (*b*) False

56. At point of projectile centripetal acceleration of object is g cos θ.

(*a*) True (*b*) False

57. Total change in momentum(2 mu sin θ) in projectile motion.

(*a*) True (*b*) False

58. Average velocity in projectile motion between any two point is u cosθ.

(*a*) True (*b*) False

59. Horizontal component of projectile motion is uniform and y^{th} component is nonuniform motion

(*a*) True (*b*) False

60. Path of on projectile with respect to another projectile is always straight line.

(*a*) True (*b*) False

61. The maximum vertical height to which a man can throw a ball is 136 m The maximum horizontal distance unto which he can throw the same ball is

(*a*) 136 m (*b*) 272 m

(*c*) 68 m (*d*) 192 m

62. A projectile is projected at 60° from vertical with initial velocity 40ms^{-1}. The velocity of projectile at t = 2s from the start will be:

(*a*) Zero (*b*) $20\sqrt{3}$ m / s

(*c*) 20 m/s (*d*) $40\sqrt{3}$m/s

63. A ball is projected from the ground with a speed 15 m /s at an angle θ with horizontal so that its range and maximum height are equal, then 'sin θ' 'will be equal to :

(*a*) 1/4 (*b*) 4

(*c*) $4/\sqrt{17}$ (*d*) 1/2

64. A projectile is projected with velocity of 25 m/s at an angle θ with the horizontal. After t seconds its inclination with horizontal becomes zero. If R represent horizontal range of projectile, the value of θ will be:

(*a*) $\frac{1}{2}\sin^{-1}\left(\frac{5t^2}{4R}\right)$ (*b*) $\tan^{-1}\left(\frac{4t^2}{5R}\right)$

(*c*) $\frac{1}{2}\sin^{-1}\left(\frac{4R}{5t^2}\right)$ (*d*) $\cot^{-1}\left(\frac{R}{20t^2}\right)$

65. The ranges and heights for two projectiles projected with the same initial velocity at angles 42° and 48° with horizontal are R_1, R_2 and H_1, H_2 respectively. Choose the correct option :

(*a*) $R_1 > R_2$ and $H_1 = H_2$ (*b*) $R_1 = R_2$ and $H_1 < H_2$

(*c*) $R_1 < R_2$ and $H_1 < H_2$ (*d*) $R_1 = R_2$ and $H_1 = H_2$

66. Average velocity of a particle in projectile motion between its starting point and the highest point of its trajectory is (projection speed = u , angle projection from horizontal = θ)

(*a*) $\frac{u}{2}\sqrt{1+2\cos^2\theta}$ (*b*) $\frac{u}{2}\sqrt{1-2\cos^2\theta}$

(*c*) $\frac{u}{2}\sqrt{1+3\cos^2\theta}$ (*d*) $u\cos\theta$

67. A particle is projected vertically upward from ground with initial velocity u such that it clears the top of a pole of height h after time t_1 in its path. It takes further time t_2 to reach the ground

(A)	u	(i)	$\frac{(t_1+t_2)^2}{8}g$
(B)	h	(ii)	$(t_{1+}t_2)$
(C)	Time of flight	(iii)	$\frac{1}{2}g(t_1+t_2)$
(D)	Maximum height	(iv)	$\frac{1}{2}gt_1t_2$

(*a*) (A)-(ii); (B)-(i); (C)-(iii); (D)-(iv)

(*b*) (A)-(i); (B)-(ii); (C)-(iii) (D)-(iv)

(*c*) (A)-(iii); (B)-(iv);(C)-(ii) (D)-(i)

(*d*) (A)-(iv); (B)-(iii); (C)-(ii); (D)-(1)

68. For ground to ground projectile motion with intitial velocity u at angle θ from horizontal

(A)	Change in speed from initial to final point	(i)	u-u cos θ
(B)	Magnitude of change in velocity from initial to final point	(ii)	u sin θ
(C)	Change in speed from initial to top	(iii)	2u sin θ
(D)	Magnitude of change in velocity from initial to top	(iv)	zero

(*a*) (A)-(i); (B)-(ii); (C)-(iii); (D)-(iv)

(*b*) (A)-(iv); (B)-(ii); (C)-(i); (D)-(iii)

(*c*) (A)-(iv); (B)-(iii); (C)-(1); (D)-(ii)

(*d*) (A)-(iv); (B)-(iii); (C)-(ii); (D)-(i)

69. A projectile is fired at an angle of θ with the horizontal. What will be the elevation angle α of the projectile at its heighest point as seen form the point of projection?

Horizontal Projectile Motion

70. When a particle is thrown horizontally, with initial velocity 'u' the resultant velocity of the projectile at any time t is given by:

(*a*) gt (*b*) $\frac{1}{2}gt^2$

(*c*) $\sqrt{u^2+g^2t^2}$ (*d*) $4\sqrt{u^2-g^2t^2}$

71. A bomber is flying horizontally with a constant speed of 15 m/s at a height of 78.4 m . The pilot has to drop a bomb at the enemy target. At what horizontal distance from the target should he release the bomb:

(*a*) Zero (*b*) 30 m (*c*) 60 m (*d*) 75 m

72. A body is thrown horizontally from the top of a tower of height 5 m. It touches the ground at a distance of 10 m from the foot of the tower. The initial velocity of the body is (g = 10 m/s^2)

(*a*) 2.5 ms^{-1} (*b*) 5 ms^{-1} (*c*) 10 ms^{-1} (*d*) 20 ms^{-1}

73. A body is thrown horizontally with a velocity $\sqrt{2gh}$ from the top of a tower of height h. It strikes the level ground through the foot of the tower at a distance x from the tower. The value of x is:

(*a*) h (*b*) $h/2$ (*c*) $2h$ (*d*) $2h/3$

74. Ball is projected with 30 m/s in horizontal direction from some height. Find time when it is 45° from horizontal.

75. A ball is projected upwards from the top of the tower with a velocity 50 ms^{-1} making an angle 30° with the horizontal. The height of tower is 70 m. After how many seconds the ball will strike the ground?

(*a*) 3 s (*b*) 5 s (*c*) 7 s (*d*) 9 s

Direction for questions from 82-85

A body x is dropped from the top of a tower. At the same time, another body y is thrown horizontally from the same position with a velocity U.

76. Both bodies will reach the ground at same time.

(*a*) True (*b*) False

77. A particle is projected from height h with horizontal velocity u, then range of particle is $u\sqrt{\frac{2H}{g}}$

(*a*) True (*b*) False

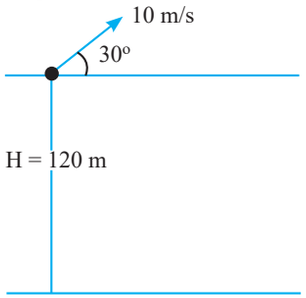
78. At the highest point of the path of projectile speed is zero.

(*a*) True (*b*) False

79. Horizontal velocity in projectile motion is zero at the highest point

(*a*) True (*b*) False

80. Find time of flight.



(*a*) 3 sec (*b*) 6 sec

(*c*) 4 sec (*d*) 7 sec

81. **Assertion (A):** Two bodies of different masses are projected horizontally with different speeds, they reach the ground simultaneously.

Reason (R): For both bodies, the vertical component of initial velocity is zero.

(*a*) Both the Assertion (A) and Reason (R) are correct, and the Reason (R) is the correct explanation of the Assertion (A).

(*b*) Both the Assertion (A) and Reason (R) are correct, but the Reason (R) is not the correct explanation of the Assertion (A).

(*c*) The Assertion (A) is correct, but the Reason (R) is incorrect.

(*d*) The Assertion (A) is incorrect, but the Reason (R) is correct.

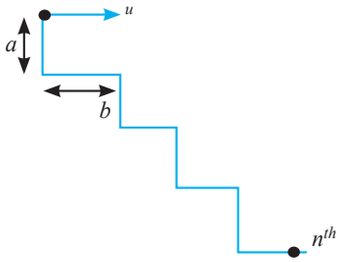
82. A bomber plane moves horizontally with a speed of 500 m/s and a bomb released from it, strikes the ground in 10 sec. Angle at which it strikes the ground will be (g = 10 m/s^2)

(*a*) $\tan^{-1}\left(\frac{1}{5}\right)$ (*b*) $\tan\left(\frac{1}{5}\right)$

(*c*) $\tan^{-1}(1)$ (*d*) $\tan^{-1}(5)$

83. Ball is Projected in Horizontal direction with speed u then find its speed when it is at a point where horizontal and vertical displacement is same.

84. Find velocity so that ball will fall on n^{th} step



Relative Motion in a Plane

85. Velocity of Ramlal $\vec{V}_R = -3\hat{i}+4\hat{j}$ and velocity of Pinky $\vec{V}_P = 4\hat{i}+3\hat{j}$ then find velocity of Ramlal with respect to Pinky.

86. Suppose you are riding a bike with a speed of 10 m /s due east relative to a person A who is walking on ground towards east. If your friend B walking on the ground due west measures your speed as 15m/ s , find the relative velocity between two reference frames A and B.

(a) Velocity of A w.r.t. B will be 5m /s due west.

(b) Velocity of B w.r.t. A will be 5m /s due east.

(c) Velocity of A w.r.t. B will be 25m /s due west.

(d) Velocity of A w.r.t. B will be 5m /s due east.

87. Ship is travelling due east at 10km/h. A ship heading 30° east of north is always due north from the first ship. The speed of the second ship in km / h is -

(a) $20\sqrt{2}$ (b) $20\sqrt{3/2}$ (c) 20 (d) $20/\sqrt{2}$

88. Car is moving with 30 m/s along east and truck is moving with speed 40 m/s at 30° north of east w.r.t. car then find velocity of truck.

89. A man 'A' moves in the north direction with a speed 10 m/s and another man B moves in E-30° – N with 10 m/s find the relative velocity of B w.r.t. A.

90. A bird is flying with a speed of 40km / hr. in the north direction. A train is moving with a speed of 40km / hr. in the west direction. A passenger sitting in the train will see the bird moving with velocity

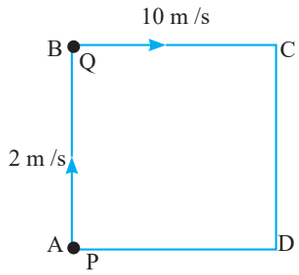
(a) 40 km/ hr in NE direction

(b) $40\sqrt{2}$ km / hr in NE direction

(c) 40 km/ hr in NW direction

(d) $40\sqrt{2}$ km / hr in NW direction

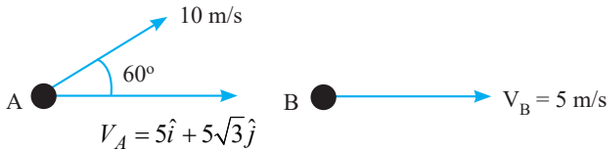
91. Two men P & Q are standing at corners A & B of square $ABCD$ of side 8 m. They start moving along the track with constant speed 2 m/s and 10m/s respectively. Find the time when they will meet for the first time.



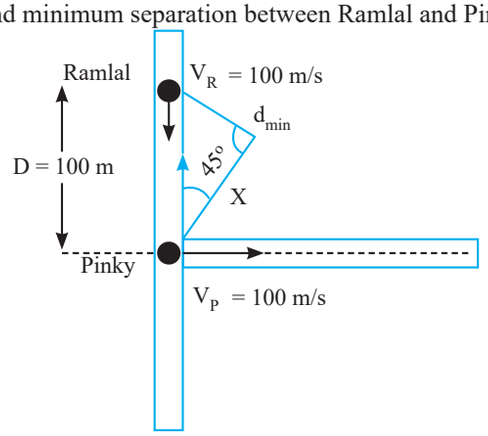
(a) 2 s (b) 3 s

(c) 1 s (d) 6 s

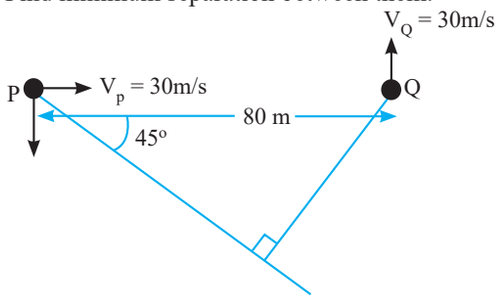
92. Find velocity of A with respect to B.



93. Find minimum separation between Ramlal and Pinky.



94. Find minimum separation between them.



95. Two particles A and B moving in $x-y$ plane are at origin at t = 0 sec. The initial velocity vectors of A and B are $\vec{u}_A = 8\hat{i}$ m / s and $\vec{u}_B = 8\hat{j}$ m / s . The acceleration of A and B are constant and are $\vec{a}_A = -2\hat{i}$ m / s^2 and $\vec{a}_B = -2\hat{i}$ m / s^2 . Column-I gives certain statements regarding particle A and B Column-I gives corresponding results. Match the statements in Column-I with corresponding results in Column-II.

Column-I		Column-II	
i	The time (in sec.) at which velocity of A relative to B is zero	A.	$16\sqrt{2}$
ii	The distance (in metres) between A and B when their relative velocity is zero	B.	$8\sqrt{2}$
iii	The time (in sec.) after $t = 0$ sec, at which A and B are at same position	C.	8
iv	The magnitude of relative velocity of A and B at instant when they are at same position	D.	4 sec

River-Swimmer Problem

96. A river is flowing from east to west at a speed of 5m/min. A man on south bank of river, capable of swimming 10m/min. in still water, wants to swim across the river in shortest time should swim

(a) Due north

(b) Due north-east

(c) Due north-east with double the speed of river

(d) None of the above

97. A boat is sailing at a velocity $\left(3\hat{i}+4\hat{j}\right)$ with respect to ground and water in river is flowing with a velocity $\left(-3\hat{i}-4\hat{j}\right)$ Relative velocity of the boat with respect to water is

(*a*) $8\hat{j}$ (*b*) $5\sqrt{2}$

(*c*) $6\hat{i}+8\hat{j}$ (*d*) $-6\hat{i}-8\hat{j}$

98. A boat takes 2 hours to go 8 km and come back in still water lake. With water velocity of 4 km / hr, the time taken for going upstream of 8 km and coming back is :-

(*a*) 140 min (*b*) 150 min

(*c*) 160 min (*d*) 60 min

99. A boat, which has a speed of 5 km/h in still water, crosses a river of width 1 km along the shortest possible path in 15 minutes. The velocity of the river water in km/h is

(*a*) 1 (*b*) 3

(*c*) 4 (*d*) $\sqrt{41}$

100. A man wishes to swim across a river 0.5 km . wide. If he can swim at the rate of 2 km /h. in still water and the river flows at the rate of 1 km /h. The angle (w.r.t. the flow of the river) along which he should swim so as to reach a point exactly opposite his starting point, should be

(*a*) 60º (*b*) 120º

(*c*) 145º (*d*) 90º

101. A man is crossing a river flowing with velocity of 5 m/s. He reaches a point directly across at distance of 60 m in 5 s. His velocity in still water should be:

(*a*) 12 m/s (*b*) 13 m/s

(*c*) 5 m/s (*d*) 10 m/s

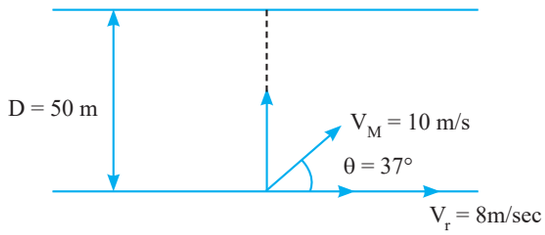
102. A person, reaches a point directly opposite on the other bank of a flowing river, while swimming at a speed of 5 m/s at an angle of 120º with the flow. The speed of the flow must be

(*a*) 2.5 m/s (*b*) 3 m/s

(*c*) 4 m/s (*d*) 1.5 m/s

103. River is flowing with speed 20 m/s a man can swim in flowing river with speed 10 m/s then find drift in a case of minimum time while the width of river is 60 m.

104. Find drift and time taken to cross the river.



105. Flow of rives is 20 m/s and Man can swim in rives with speed 10 m/s, then find angle at Which man have to swim to counter the river's flow and reach the exact opposite point with zero drift, following the minimum path?

(*a*) 30º (*b*) 60°

(*c*) 120° (*d*) None of these

106. Flow of river is 200 m/s and man can swim in river with speed 10m/s then find angle at which man have to swin for minimum time

(*a*) 30º (*b*) 90°

(*c*) 120° (*d*) None of these

107. A boat moving towards east with velocity 4m/s with respect to still water and river is flowing towards north with velocity 2 m/s and the wind is blowing towards north with velocity 6 m/s. The direction of the flag blown over by the wind hoisted on the boat is

(*a*) North - west (*b*) south-east

(*c*) $\tan^{-1}(1/2)$ with east (*d*) North

108. A river flows with a speed more than the maximum speed with which a person can swim in still water. He intends to cross the river by the shortest possible path [*i.e.* he wants to reach the point on the opposite bank which directly opposite to starting point]. Which of the following is correct?

(*a*) He should start normal to river bank.

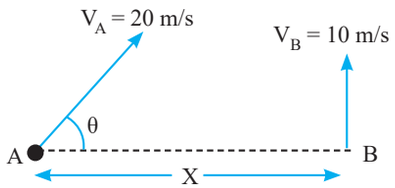
(*b*) He should start in such away that he moves normal to bank, relative to bank.

(*c*) He should start in particular calculated direction making an obtuse angle with downstream.

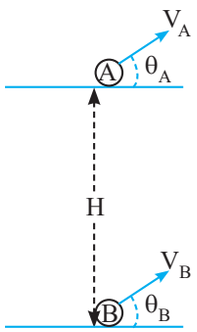
(*d*) The man cannot cross the river in that way.

Condition of Collision

109. Two particles A and B are projected from the ground simultaneously in the directions shown in the figure with initial velocities V_A = 20 m /s and V_B = 10 m/s respectively. They collide after 0.5 s. Find out the angle θ and the distance *x*.



110. Two balls are projected as shown in figure, find time of collision and condition of collision?



111. Two particles having position vectors $\vec{r}_1 = (3\hat{i} + 5\hat{j})$ meters and $\vec{r}_2 = (-5\hat{i} - 3\hat{j})$ meres are moving with velocities $\vec{v}_1 = (2\hat{i} + 3\hat{j})$ m/s and $v_2 = (\alpha\hat{i} + 7\hat{j})$ m/s. If they collide after 2 seconds, the value of 'α' is:

(*a*) 2 (*b*) 4

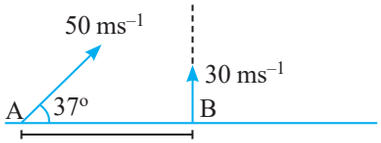
(*c*) 6 (*d*) 8

112. Two boys are standing at the ends A and B of a ground where AB = *a*. The boy at B starts running in a direction perpendicular to AB with velocity v_1. The boy at A starts running simultaneously with velocity *v* and catches the other boy in a time *t*, where *t* is-

(*a*) $\frac{a}{\sqrt{v^2 + v_1^2}}$ (*b*) $\frac{a}{\sqrt{v^2 - v_1^2}}$

(*c*) $\frac{a}{(v - v_1)}$ (*d*) $\frac{a}{(v + v_1)}$

113. Balls A and B are thrown from two points lying on the same horizontal plane separated by a distance of 120 m. Which of the following statements is correct?



(*a*) The balls can never meet

(*b*) The balls can meet if the ball B is thrown 1 s later

(*c*) The two balls meet at a height of 45 m

(*d*) None of the above

114. Four persons P, Q, R and S are initially at the four corners of a square of side *d*. Each person now moves with a constant speed *v* in such a way that P always moves directly towards Q, Q towards R, R towards S, and S towards P. The four persons will meet after time

(*a*) $\frac{d}{2v}$ (*b*) $\frac{d}{v}$

(*c*) $\frac{3d}{2v}$ (*d*) The will never meet

115. A 2m wide truck is moving with a uniform speed V_o = 8 ms^{-1} along a straight horizontal road. A man starts to cross the road with a uniform speed *v*, when the truck is 4 m away from him. The minimum value of *v* so that he can cross the road safely is

(*a*) 2.62 ms^{-1} (*b*) 4.6 m^{-1}

(*c*) 3.57 ms^{-1} (*d*) 1.414 ms^{-1}

Rain-Man Problem

116. Rain is falling vertically with a speed of 30 ms^{-1}. A woman rides a bicycle with a speed of 10 ms^{-1}. in the north direction. What is the direction in which she should hold her umbrella?

117. A man standing on a road has to hold his umbrella at 30° with the vertical to keep the rain away. He throws the umbrella and starts running at 10 km/hr then he finds that rain drops are hitting his head vertically, then speed of rain drops with respect to moving man:-

(*a*) 20 km/hr (*b*) $10\sqrt{3}$ km/hr

(*c*) $10\sqrt{3}$ km/hr (*d*) 10 km/hr

118. A man walks in rain with a velocity of 5 kmh^{-1}. The rain drops strike at him at an angle of 45° with the horizontal. Velocity of rain if it is falling vertically downward-

(*a*) 5 kmh^{-1} (*b*) 4 kmh^{-1}

(*c*) 3 kmh^{-1} (*d*) 1 kmh^{-1}

119. A car with vertical windshield moves in a rain storm at a speed of 40 km/hr. The rain drops fall vertically with constant speed of 20 m/s. The angle at which rain drops strike the windshield is

(*a*) $\tan^{-1}\frac{5}{9}$ (*b*) $\tan^{-1}\frac{9}{95}$

(*c*) $\tan^{-1}\frac{3}{2}$ (*d*) $\tan^{-1}\frac{2}{3}$

120. A man sitting in a bus travelling in a direction from west to east with a speed of 40 km/h observes that the rain-drops are falling vertically downwards. To another man standing on ground the rain will appear.

(*a*) To fall vertically downwards

(*b*) To fall at an angle going from west to east

(*c*) To fall at an angle going from east to west

(*d*) The information given is insufficient to decide the direction of the rain

Kinematics of Circular Motion

121. Find angular and linear displacement in one revolution.

122. Find angle rotated by angular displacement vector in one rotation.

123. Find angular displacement of hr. hand, minute hand and second hand of clock in 12 hr.

124. A fly wheel is accelerated uniformly from rest and rotates through 5 rad in the first second. The angle rotated by the fly wheel in the next second, will be:

(*a*) 7.5 rad (*b*) 15 rad

(*c*) 20 rad (*d*) 30 rad

125. Angular velocity of minute hand of a clock is:-

(*a*) $\frac{2\pi}{1800}$ rad/s (*b*) 8π rad/s

(*c*) $\frac{\pi}{1800}$ rad/s (*d*) $\frac{\pi}{30}$ rad/s

126. Angular displacement of object θ = t^2 + 2*t* + 5 then, find angular speed at 2 sec and in 2-sec.?

127. Angular velocity of a rotating object is given by the equation: $\omega = 6t - t^2 + 6$ Find the time at which the angular acceleration becomes zero?

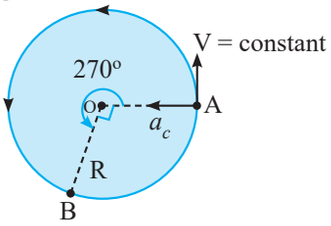
128. Object completes 7 rotation in 22 sec on circular path of radius 1 m then find angular speed.??

129. A body performing uniform circular motion completed 140 revolution in a second. Its angular speed is

(*a*) 880 rad/s (*b*) 440 rad/s

(*c*) 220 rad/s (*d*) 240 rad/s

130. Find average acceleration between A and B.



131. Find angular speed of hr. hand

132. Body moving in uniform circular motion rotated by 120° then find change in velocity.

133. In uniform circular motion acceleration is:?

(*a*) Constant (*b*) Variable

134. An object moving in a circular path at constant speed has constant

(*a*) Energy (*b*) Velocity

(*c*) Acceleration (*d*) Displacement

135. The angle between velocity vector and acceleration vector in uniform circular motion is

(*a*) 0° (*b*) 180° (*c*) 90° (*d*) 45°

136. Speed of an object moving in circular path of radius 10 m with angular speed 2 rad/s is

(*a*) 10 m/s (*b*) 5 m/s (*c*) 20 m/s (*d*) 30 m/s

137. Centripetal acceleration of a cyclist completing acceleration of a cyclist completing 7 rounds in a minute along a circular track of radius 5 m with a constant speed, is

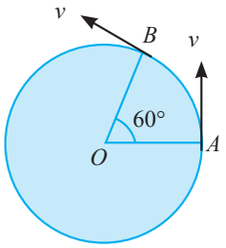
(*a*) 2.7 m/s^2 (*b*) 4 m/s^2 (*c*) 3.78 m/s^2 (*d*) 6 m/s^2

138. If the frequency of an object in uniform circular motion is doubled, its acceleration becomes

(*a*) Two times (*b*) Four times

(*c*) Half (*d*) One fourth

139. A particles is moving in a circle of radius *r* having centre at O, with a constant speed *v*. The magnitude of change in velocity in moving from A to B is



(*a*) $2v$ (*b*) 0 (*c*) $\sqrt{3}v$ (*d*) v

140. A particle is moving with constant speed in a circular path. When the particle turns by an angle 90°, the ratio of instantaneous velocity is $\pi : x\sqrt{2}$. The value of *x* will be

(*a*) 2 (*b*) 5 (*c*) 1 (*d*) 7

141. A body revolves with constant speed *v* in a circular path of radius *r*. The magnitude of its average acceleration during motion between two points in diametrically opposite direction is

(*a*) 0 (*b*) $\frac{v^2}{r}$

(*c*) $\frac{2v^2}{\pi r}$ (*d*) $\frac{v^2}{2r}$

142. A body is moving on a circle of radius 80 m with a speed 20 m/s which is decreasing at the rate 5 m/s^2 at an instant. The angle made by its acceleration with its velocity is

(*a*) 45° (*b*) 90° (*c*) 135° (*d*) 0°

143. A car is moving at a speed of 40 m/s on a circular track of radius 400 m. This speed is increasing at the rate of 3 m/s^2. The acceleration of car is

(*a*) 4 m/s^2 (*b*) 7 m/s^2

(*c*) 5 m/s^2 (*d*) 3 m/s^2

144. A car travelling with linear velocity *v* on circular path of radius *r* its speed increasing at rate α, then find acceleration.

145. The distance of a particle moving on a circle of radius 12 m measured from a fixed point on the circle and measured along the circle is given by $s = 2t^3$ (in meters). The ratio of its tangential to centripetal acceleration at $t = 2$sec.

(*a*) 4 : 1 (*b*) 1 : 2 (*c*) 2 : 1 (*d*) 3 : 1

146. A car is going round a circle of radius R_1 with constant speed. Another car is going round a circle or radius R_2 with constant speed. If both of them take same time to complete the circles, the ratio of their angular speeds and linear speeds will be

(*a*) $\sqrt{\frac{R_1}{R_2}}, \frac{R_1}{R_2}$ (*b*) 1, 1

(*c*) $1, \frac{R_1}{R_2}$ (*d*) $\frac{R_1}{R_2}$

147. If θ is angle between the velocity and acceleration of a particle moving on a circular path with decreasing speed, then

(*a*) $\theta = 90°$ (*b*) $0° < \theta < 90°$

(*c*) $90° < \theta < 180°$ (*d*) $0° \le \theta \le 180°$

148. If speed of an object revolving in a circular path is doubled angular speed is reduced to half of original value, then centripetal acceleration will become/remain

(*a*) Same (*b*) Double

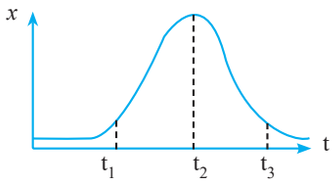
(*c*) Half (*d*) Quadruple

MR* CORNER

1. **Statement I:** When a particle moves with constant velocity, its average velocity, it's instantaneous velocity and it's speed are all equal.

   Statement II: If a particle is accelerating, it is either speeding up or speeding down.

2. Given figure shows the x-coordinate of particle is function of time.



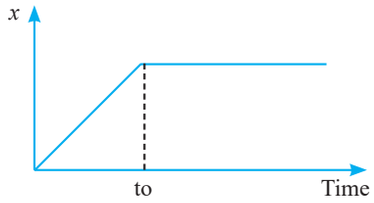
   (*i*) Sign of velocity at time t_1 = _____.

   (*ii*) Sign of acceleration at time t_2 = _____.

   (*iii*) Sign of velocity at time t_3 = _____.

   (*iv*) sign of acceleration at time t_3 = _____.

3. For a given figure



   Statement I: The velocity increases up to a time to, and then becomes constant.

   Statement II: The particle moves at a constant velocity up to a time to, and then stops.

4. A stone is released from an elevator going up with an acceleration *a* The acceleration of the stone before and after the release is _____ and _____ respectively.

5. *A* person standing near the edge of the top of a building has two balls *A* and *B*. The ball *A* is thrown vertically upward and *B* is released at the same time. If ball *A* hits ground with speed V_A and ball *B* hits ground with speed V_s then we have

   (*a*) $V_A = V_B$

   (*b*) $V_A > V_B$

   (*c*) $V_B > V_A$

   (*d*) Relation between V_A & V_B depends upon height of building

6. Two bullets are fired simultaneously, horizontally and with different speeds from the same place. Then which of the following statements is correct.

   (*a*) Both bullet will hit the ground simultaneously and with some speed

   (*b*) Bullet with faster speed will hit ground first.

   (*c*) Both bullet will hit ground simultaneously, will have different vertical component of velocity while hitting ground.

   (*d*) Both bullet will hit ground simultaneously, will have some vertical component of velocity but they will hit ground with different net speed.

7. **Statement I:** It is possible to have a situation in which $\left|\frac{d\vec{v}}{dt} \neq 0\right|$ but $\frac{d|\vec{v}|}{dt} = 0$.

   Statement II: The average velocity of particle moving on a straight line is zero in time interval. It is possible that the instantaneous velocity is never zero in the interval.

8. An object may have_____Without having_____.

   (*a*) Varying speed, varying velocity.

   (*b*) Varying velocity, varying speed.

   (*c*) Non zero acceleration, varying velocity.

   (*d*) Both (*b*) and (*c*)

9. **Statement I:** For motion in straight line, if the position and velocity have same sign, the particle is moving towards the origin.

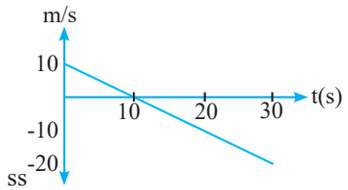
   Statement II: If the velocity is zero for a time interval, the acceleration in zero at any instant within the time interval.

   Statement III: The velocity of a particle is zero at t = 0, then acceleration at t = 0 must be zero.

10. **Statement I:** The magnitude of average velocity in an interval is always equal to its average speed in that interval.

    Statement II: If is not impossible to have a situation in which is which the speed of the particle is never zero but the average speed in an interval is zero.

11. For given graph



    Statement I: The particle has never turned around.

    Statement II: The average speed in the interval 0 to 10 seconds is the same as average speed in interval 10s to 20s.

12. **Statement I:** If the acceleration of a particle as seen from two frames S_1 and S_2 have equal magnitude of 5 m/s², then frames must be at rest with respect to each other.

    Statement II: for above given case, acceleration of S_2 with respect to S_1 May be anything between zero and 10 m/s²

Answer Key

1. (i) $\sqrt{2}$ m/s, (ii) 0 m/s **2.** $\vec{v} = 6\hat{i} + 4\hat{j}, \vec{a} = 6\hat{i} + -2\hat{j}$ **3.** $\vec{a} = -4\text{m/s}^2$ **4.** $5\sqrt{2}$ m/s **5.** $\frac{\sqrt{5}}{4}$

6. $\theta = 60°$ **7.** $20\sqrt{2}$ **8.** 1 s **9.** (*a*) **10.** (60) **11.** $\vec{a} = \frac{5}{2}\hat{i} + \frac{5}{2}\hat{j}$ **12.** (*b*) **13.** $\sqrt{5}$

14. $(4\hat{i} - 4\hat{j})\,m/s, (12\hat{i} + 6\hat{i})m$ **15.** A → III, B → IV, C → II **16.** (*a*) **17.** (*d*) **18.** $y = x^2$

19. $x^2 + y^2 = 3^2$ (Circle) **20.** $\frac{x^2}{16} + \frac{y^2}{9} = 1$ (ellipse) **21.** (*d*) **22.** $y = \frac{x^2}{4}$ **23.** $x^2 - y^2 = K$ **24.** (*b*)

25. $\mu = 45\text{m}, T = 6\text{s}, R = 240$ m, $\theta = 37°$ **26.** (*d*) **27.** (*c*) **28.** (*b*) **29.** (*b*) **30.** (*b*) **31.** (*a*)

32. (*b*) **33.** (*d*) **34.** $R = \frac{1}{2} g\, t_1 . t_2$ **35.** (*d*) **36.** (*c*) **37.** (*a*) **38.** (*b*) **39.** (*c*) **40.** (*c*)

41. $\theta = 53, T_f = 8, R = 240\text{m}$ **42.** $\theta = \tan^{-1}(\beta/\alpha), R = \frac{2\alpha\beta}{r}, H = \frac{\beta^2}{2r}$ **43.** $t_f = 2\text{sec}, H = 2\text{m}, R = 6\text{m}$

44. $h = \frac{x_1 x_2 \tan\theta}{x_1 + x_2}$ **45.** $h = \frac{10}{7} m$ **46.** $t = \frac{u}{g\sin\theta}$ **47.** (*a*) **48.** (*c*) **49.** (*a*) **50.** (*a*)

51. (*a*) **52.** (*b*) **53.** (*a*) **54.** (*a*) **55.** (*b*) **56.** (*a*) **57.** (*a*) **58.** (*b*) **59.** (*a*) **60.** (*a*)

61. (*b*) **62.** (*b*) **63.** (*c*) **64.** (*d*) **65.** (*b*) **66.** (*c*)

67. (*c*) **68.** (*c*) **69.** $\tan\alpha = \frac{\tan\theta}{2}$ **70.** (*c*) **71.** (*c*) **72.** (*c*) **73.** (*c*) **74.** $t = 3$ sec.

75. (*c*) **76.** (*a*) **77.** (*a*) **78.** (*b*) **79.** (*b*) **80.** (*b*) **81.** (*a*) **82.** (*a*)

83. $|u| = (\sqrt{5})u$ **84.** $u = \sqrt{\frac{nb^2 g}{2a}}$ **85.** $\vec{V}_{RP} = -7\hat{i} + \hat{j}$ **86.** (*d*) **87.** (*c*) **88.** $\vec{u}_\tau = (20\sqrt{3} + 30)\hat{i} + 20\hat{j}$

89. 10 m/s **90.** (*b*) **91.** (*b*) **92.** $\vec{U}_{AB} = 5\sqrt{3}\hat{j}$ **93.** $D_{min} = 50\sqrt{2}$ m **94.** $40\sqrt{2}$ m

95. (i) –D, (ii) –A, (iii) –C, (iv) –B **96.** (*a*) **97.** (*c*) **98.** (*c*) **99.** (*b*) **100.** (*b*) **101.** (*b*) **102.** (*a*)

103. 120 m **104.** 8.33 sec, Drift = $(8 + V_m \cos 37°)$ time **105.** (*d*) **106.** (*b*) **107.** (*a*) **108.** (*d*) **109.**

$30°, x = 5\sqrt{3}$ m **110.** $t = \frac{H}{V_B \sin\theta_B - V_A \sin\theta_A}, V_A \cos\theta_A = V_B \cos\theta_B$ **111.** (*c*) **112.** (*b*) **113.** (*c*)

114. (*b*) **115.** (*c*) **116.** $\theta = \tan^{-1}\left(\frac{1}{3}\right)$ from vertical **117.** (*b*) **118.** (*a*) **119.** (*a*)

120. (*b*) **121.** $\theta = 2\pi$ radian, linear displacement = 0 **122.** (0°) **123.** $\theta_{\text{Hr. hand}} = 2\pi$ rad, $\theta_{\text{Mn. hand}} = 24\pi$ rad, $\theta_{\text{Sec. hand}} = 1440\pi$ rad

124. (*b*) **125.** (*c*) **126.** 6 rad/s **127.** t = 3 sec, 4rad/sec **128.** 2 rad/s **129.** (*a*) **130.** $\frac{4v^2}{3\sqrt{2}\pi R}$ **131.** $\frac{2\pi}{60}$ rad/s

132. $v\sqrt{3}$ m/s **133.** (*b*) **134.** (*a*) **135.** (*c*) **136.** (*c*) **137.** (*c*) **138.** (*b*)

139. (*d*) **140.** (*a*) **141.** (*c*) **142.** (*c*) **143.** (*c*) **144.** $\sqrt{\left(\frac{u^2}{r}\right)^2 + \alpha^2}$ **145.** (*b*) **146.** (*c*) **147.** (*c*)

148. (*a*)

MR* CORNER

1. I. True, II. False **2.** (i) Positive, (ii) Negative, (iii) Negative, (iv) Positive **3.** I. False, II. True
4. *a* upwards, g downwards **5.** (*b*) **6.** (*d*) **7.** I. True, II. False **8.** (*d*)
9. I. False, II. True, III. False **10.** Both are False. **11.** I. False, II. True **12.** I. False, II. True